

BIO 004 · HUMAN ANATOMY · SOLANO COMMUNITY COLLEGE

Lab Structure List

Systems reference, organized in textbook chapter order

This is your master lab study guide. Because this course has no assigned textbook, the list is organized by body system in the order a standard human anatomy textbook follows, so it doubles as your chapter-by-chapter reference. These are the lab structures and concepts to learn for the lab portion of each exam. Be prepared to identify structures on models, images, the cadaver, and microscope slides. **This is an extensive master list.** It would be impossible to test everything, but everything here is testable. Cross-reference it with your lab worksheets; occasionally a worksheet will add or omit a structure, so this master list is the authority. Arteries, veins, and nerves are grouped into their own sections by region (not repeated within each muscle group). Your lab sprints will sequence these structures by when they are due.

1 · Orientation and the Body Plan

1. **Regional landmarks:** all anterior and posterior body regional terms (cervical, buccal, cranial, carpal, sternal, and so on).
2. **Directional terms:** anterior/posterior, medial/lateral, cranial (cephalad)/caudal, ventral/dorsal, proximal/distal, superficial, deep. Be able to use them in a sentence (the trachea is ventral to the spine).
3. **Planes and sections:** midline, median (midsagittal), parasagittal, coronal (frontal), transverse (horizontal/cross-sectional), oblique.
4. **Body cavities:** dorsal/ventral, cranial, vertebral, thoracic, abdominopelvic, mediastinum. Know the structure that separates the thoracic from the abdominopelvic cavity (diaphragm).
5. **Mediastinum:** structures forming the superior, inferior, ventral, dorsal, and lateral borders; the contents.
6. **Retroperitoneum:** the retroperitoneal structures (memory device: SADPUCKER).
7. **Serous membranes and cavities:** pericardium, peritoneum, and pleura, each with parietal layer, visceral layer, cavity, and fluid.
8. **Abdominopelvic quadrants** (RUQ, LUQ, RLQ, LLQ) and the major organs in each.
9. **Microscope:** the parts of the microscope and how to calculate total magnification.

2 · Tissues (Histology)

Epithelial tissue

1. **General structures:** apical surface, lateral surface, basal surface, basement membrane, lamina propria (when present), nuclei, lumen/free surface, cilia, microvilli, goblet cells, keratin (each when present).
2. **Identify the epithelia** (general function and location): simple squamous; keratinized stratified squamous; non-keratinized stratified squamous; simple cuboidal; stratified cuboidal; simple columnar; pseudostratified ciliated columnar; transitional.
3. Know which tissues carry cilia, microvilli, and goblet cells, and the function of each.

Connective tissue

1. **Connective tissue proper** (identify tissue, the listed structures, function, location): loose areolar (collagen fibers, elastic fibers, fibroblasts, mast cells); adipose (adipocytes, fat vacuoles, blood vessels); reticular (reticular fibers); dense regular (collagen fibers, fibroblasts); dense irregular (collagen fibers, fibroblasts); elastic (elastin fibers).
2. **Cartilage**: hyaline (perichondrium, lacunae, chondrocytes, matrix); elastic (same plus elastin fibers); fibrocartilage (lacunae, chondrocytes, matrix, collagen, no perichondrium).
3. **Osseous tissue**: osteons, central (Haversian) canal, lacunae, lamellae, osteocytes, canaliculi.

Muscle tissue

1. **Skeletal**: histology (dark/light striations of actin and myosin, nuclei, sarcolemma); sarcomere (Z disc, A band, I band, H zone, M line, actin, myosin); general (endomysium, perimysium, epimysium, muscle fiber, fascicle, muscle belly, tendon). Striated, voluntary.
2. **Cardiac**: intercalated discs (know the two cell junctions there and the function of each). Striated, involuntary.
3. **Smooth**: non-striated, involuntary.

Nervous tissue

1. **Multipolar neuron**: cell body (soma), nucleus, nucleolus, dendrites, axon, axon hillock, synaptic bulb (terminal end).
2. **Spinal cord (slide)**: white matter, grey matter, ventral horn, dorsal horn, lateral horn (when present).
3. **Cerebral cortex**: pyramidal cells. **Cerebellum**: Purkinje cells.
4. **Nerve (cross section)**: axon, Schwann cells, nodes of Ranvier, epineurium, perineurium, endoneurium.

3 · Integumentary System

1. **Epidermis layers** (deep to superficial): stratum basale, stratum spinosum, stratum granulosum, stratum lucidum (thick skin only), stratum corneum. Know thick from thin skin (which layer is missing in thin skin).
2. **Dermis**: papillary layer, reticular layer, areolar CT, dense irregular CT, dermal papillae, eccrine sweat gland and duct, hair / follicle / root / bulb / shaft, arrector pili muscle, sebaceous gland and duct, apocrine gland and duct, Meissner's corpuscles, free nerve endings, capillary loops.
3. **Hypodermis (subcutaneous)**: adipose CT, areolar CT, lamellated (Pacinian) corpuscle.
4. The three structures within a dermal papilla: Meissner's corpuscles, free nerve endings, capillary loops.

4 · Skeletal System

Bone tissue (micro) and the long bone

1. **Osseous tissue model**: osteon, central (Haversian) canal, Volkmann's (perforating) canal, canaliculi, lacunae, osteocytes, inner and outer circumferential lamellae, interstitial lamellae, concentric lamellae, medullary cavity, periosteum (inner osteogenic, outer fibrous), Sharpey's fibers.
2. **Long bone gross**: epiphysis, metaphysis, diaphysis, medullary cavity, endosteum, periosteum (osteogenic and fibrous layers), articular cartilage, epiphyseal line (adult) / epiphyseal plate (child), compact bone, spongy bone, collagen fibers (between the lamellae), red and yellow marrow.

Axial skeleton: the skull

1. **Cranial and facial bones:** frontal, occipital, parietal, temporal, sphenoid, ethmoid, maxilla, zygomatic, nasal, lacrimal, mandible.
2. **Sutures:** coronal, sagittal, squamous (squamosal), lambdoid.
3. **Lateral view:** glabella, supraorbital foramen/notch and margin, zygomatic arch, zygomatic process of temporal bone, temporal process of zygomatic bone, mastoid process, external acoustic (auditory) meatus, styloid process, articular tubercle, alveolar process, anterior nasal spine, infraorbital foramen.
4. **Anterior view:** supraorbital foramen/notch and margin, glabella, perpendicular plate of ethmoid, vomer, middle and inferior nasal conchae and meatuses, anterior nasal spine, alveolar process, infraorbital foramen, nasion.
5. **Posterior view:** superior and inferior nuchal lines, external occipital protuberance, vomer, occipital condyles, palatine bone, palatine process of maxilla, incisive fossa/foramen, mastoid process.
6. **Base, exterior:** incisive foramen, transverse and median palatine sutures, palatine bone, vomer, medial and lateral pterygoid plates, foramen ovale, foramen spinosum, foramen lacerum, foramen rotundum, carotid canal, jugular foramen, hypoglossal canal, foramen magnum, nuchal lines, external occipital protuberance, mastoid process, articular tubercle, occipital condyles, styloid process, mandibular fossa, zygomatic arch, inferior orbital fissure, choana.
7. **Base, interior:** anterior, middle, posterior cranial fossae, frontal crest, frontal sinus, cribriform plate of ethmoid, crista galli, olfactory foramina, lesser and greater wings of sphenoid, hypophyseal fossa (sella turcica), dorsum sellae, tuberculum sellae, internal acoustic meatus, jugular foramen, internal occipital protuberance, hypoglossal canal, foramen lacerum, foramen magnum, foramen spinosum, foramen ovale, optic canal, anterior clinoid processes. (Know what sits in the hypophyseal fossa: the pituitary.)
8. **Ethmoid bone:** crista galli, perpendicular plate, cribriform plate, olfactory foramina.
9. **Sphenoid bone:** medial and lateral pterygoid plates, hypophyseal fossa (sella turcica), dorsum sellae, tuberculum sellae, greater and lesser wings, superior orbital fissure.
10. **Orbit:** sphenoid bone, lacrimal bone, lacrimal canal, optic canal, superior orbital fissure, inferior orbital fissure.
11. **Mandible:** condylar process, coronoid process, mandibular notch, mental protuberance, mental foramen, alveolar process and margin, body, ramus, mandibular foramen, angle.
12. **Hyoid bone:** body, greater cornu, lesser cornu.
13. **Nasal septum:** septal cartilage, perpendicular plate of ethmoid, vomer.
14. **Fetal skull:** anterior fontanel, posterior fontanel, paired anterolateral fontanels, paired posterolateral fontanels.

Axial skeleton: vertebral column and thoracic cage

1. **Regions:** cervical (C1-C7), thoracic (T1-T12), lumbar (L1-L5), sacrum (5 fused), coccyx (3-4 fused).
2. **Normal curves:** cervical, thoracic, lumbar, sacrococcygeal; label each primary or secondary and know the developmental milestone behind each secondary curve.
3. **Abnormal curves:** scoliosis, lordosis, kyphosis.
4. **C1 (atlas):** anterior and posterior tubercle, facet for the dens, anterior and posterior arch, superior and inferior articular facets and processes, transverse process, transverse foramen, vertebral foramen.
5. **C2 (axis):** dens, articular facets and processes, body, transverse process, transverse foramen, vertebral foramen, spinous process, vertebral arch.
6. **Typical cervical / thoracic / lumbar vertebrae:** body, pedicle, lamina, vertebral arch, vertebral foramen, transverse process, spinous process, superior and inferior articular facets and processes; transverse foramina (cervical); costal facets (thoracic). Be able to ID a specific vertebra and the region it belongs to from its features.
7. **Rib:** head, neck, shaft.

8. **Sacrum and coccyx:** sacral promontory, superior articular process and facet, anterior and posterior sacral foramina, coccyx, sacral hiatus, median and lateral sacral crests, articular surface, base, apex, sacral tuberosity.
9. **Sternum:** manubrium, sternal angle, body, xiphoid process, jugular notch, clavicular notch, costal notch.
10. **Intervertebral disc:** annulus fibrosus, nucleus pulposus.

Appendicular skeleton: pectoral girdle and upper limb

1. **Clavicle:** sternal end, acromial end, conoid tubercle.
2. **Scapula:** acromion, coracoid process, spine, infraspinous fossa, supraspinous fossa, subscapular fossa, glenoid cavity, supraglenoid and infraglenoid tubercles, superior border, superior angle, suprascapular notch, axillary (lateral) border, vertebral (medial) border, inferior angle.
3. **Humerus:** head, anatomical neck, surgical neck, greater and lesser tubercles, intertubercular (bicipital) sulcus, deltoid tuberosity, radial groove, lateral and medial epicondyles, lateral and medial supracondylar ridges, trochlea, capitulum, coronoid fossa, radial fossa, olecranon fossa.
4. **Radius:** head, neck, styloid process, radial tuberosity, ulnar notch.
5. **Ulna:** head, styloid process, ulnar tuberosity, radial notch, coronoid process, trochlear notch, olecranon.
6. **Carpals** (So Long To Pinky, Here Comes The Thumb): proximal row lateral to medial — scaphoid, lunate, triquetrum, pisiform; distal row medial to lateral — hamate, capitate, trapezoid, trapezium. **Metacarpals** 1-5 (lateral to medial). **Phalanges:** proximal, middle, distal (no middle on digit 1).

Appendicular skeleton: pelvic girdle and lower limb

1. **Coxal bone — ilium:** iliac crest, auricular surface, iliac fossa, greater and lesser sciatic notches, ala, ASIS, AIIS, PSIS, PIIS.
2. **Ischium:** ischial spine, ischial tuberosity, lesser sciatic notch, ramus. **Pubis:** body, superior and inferior rami, pubic crest, pubic tubercle. **Plus:** acetabulum, obturator foramen, sacroiliac joint, pubic symphysis.
3. **Femur:** head, fovea capitis, neck, greater and lesser trochanters, intertrochanteric line and crest, gluteal tuberosity, linea aspera (medial and lateral lips), patellar surface, adductor tubercle, medial and lateral supracondylar lines, medial and lateral epicondyles, medial and lateral condyles, intercondylar fossa.
4. **Patella.**
5. **Tibia:** medial and lateral condyles, intercondylar eminence, tibial tuberosity, fibular notch, anterior crest, medial malleolus. **Fibula:** head, lateral malleolus.
6. **Tarsals** (the three C's surround the N and T): talus, calcaneus, navicular, medial / intermediate / lateral cuneiforms, cuboid. **Metatarsals** I-V (medial to lateral). **Phalanges:** hallux has proximal and distal only; toes 2-5 have proximal, middle, distal.

Joints

1. For any tagged joint: classify as fibrous, cartilaginous, or synovial; give the synovial subtype (hinge, ball-and-socket, saddle, etc.); classify by movement (synarthrosis, amphiarthrosis, diarthrosis); and name the movements that occur there.
2. **Named joints:** atlanto-axial, atlanto-occipital, glenohumeral, acromioclavicular, sternoclavicular, manubriosternal, radioulnar, talocrural, tibiofibular, temporomandibular, sacroiliac, pubic symphysis.
3. **Typical synovial joint:** articular cartilage, synovial membrane, joint cavity with synovial fluid.
4. **Knee joint:** femur, tibia, fibula, patella, patellar ligament, quadriceps tendon, medial (tibial) and lateral (fibular) collateral ligaments, medial and lateral menisci, anterior and posterior cruciate ligaments.

5 · Muscular System

Muscle microanatomy

1. **Sarcomere:** Z disc, A band, I band, H zone, M line, actin, myosin, light chains, dark chains.
2. **General:** endomysium, perimysium, epimysium, muscle fiber, fascicle, myofibril, sarcolemma, sarcoplasm, transverse (T) tubule, nucleus, mitochondria, triad, terminal cisternae, sarcoplasmic reticulum.
3. **Neuromuscular junction:** pre- and post-synaptic neuron, soma, nucleus, nucleolus, dendrites, axon, axon hillock, glial cells (when present), lipofuscin, Nissl bodies, axon terminal, axoplasm, axolemma, synapse, synaptic end bulbs, vesicles.

For every muscle on lab exams: identify the muscle and its origin, insertion, and action. Innervation may be included and is discussed in lab.

Gross muscles by region

1. **Chest and anterior arm:** pectoralis major, pectoralis minor, serratus anterior, subscapularis, coracobrachialis, biceps brachii (long and short heads), brachialis.
2. **Anterior forearm:** superficial — pronator teres, flexor carpi ulnaris, palmaris longus; intermediate — flexor digitorum superficialis; deep — flexor digitorum profundus, flexor pollicis longus, pronator quadratus.
3. **Posterior forearm:** brachioradialis, extensor carpi radialis longus and brevis, extensor carpi ulnaris, extensor digitorum, extensor digiti minimi, abductor pollicis longus, extensor pollicis longus and brevis, extensor indicis.
4. **Posterior thorax, shoulder, posterior arm:** superficial — trapezius, latissimus dorsi, teres major, deltoid; deep — rhomboid major, rhomboid minor, levator scapulae, supraspinatus, infraspinatus, teres minor; posterior arm — triceps brachii (long, lateral, medial heads). Rotator cuff (SITS): supraspinatus, infraspinatus, teres minor, subscapularis.
5. **Neck:** platysma, sternocleidomastoid, omohyoid, sternohyoid, sternothyroid, thyrohyoid.
6. **Facial expression and mastication:** depressor and levator anguli oris, depressor labii inferioris, levator labii superioris, orbicularis oris, risorius, zygomaticus major and minor, nasalis, procerus, temporalis, buccinator, masseter, mentalis, occipitofrontalis (frontal and occipital bellies), galea aponeurotica.
7. **Abdominal wall:** rectus abdominis, external abdominal oblique, internal abdominal oblique, transversus abdominis.
8. **Anterior, medial, lateral thigh:** iliopsoas (iliacus + psoas major), quadriceps femoris (rectus femoris, vastus lateralis, vastus intermedius, vastus medialis), sartorius, gracilis, adductor magnus, adductor longus, adductor brevis, pectineus, tensor fasciae latae, iliotibial (IT) band.
9. **Gluteal region and posterior thigh:** gluteus maximus, gluteus medius, hamstrings (biceps femoris, semitendinosus, semimembranosus).
10. **Leg:** tibialis anterior, extensor hallucis longus, extensor digitorum longus, fibularis tertius, fibularis brevis, fibularis longus, tibialis posterior, flexor digitorum longus, flexor hallucis longus, gastrocnemius, soleus, plantaris.
11. **Extraocular (eye) muscles:** lateral rectus, medial rectus, superior rectus, inferior rectus, superior oblique, inferior oblique, levator palpebrae superioris.

6 · Nervous System

Spinal cord and spinal nerves

1. **Spinal cord gross:** meninges (dura mater, arachnoid mater, pia mater); enlargements (cervical enlargement, lumbar enlargement); conus medullaris, cauda equina, filum terminale.

2. **Spinal cord cross section:** ventral horn, lateral horn, dorsal horn, grey matter, white matter, grey commissure, central canal.
3. **Spinal nerve (PNS):** ventral root, dorsal root, dorsal root ganglion, spinal nerve, ventral ramus, dorsal ramus, grey and white rami communicantes, sympathetic chain ganglion. Know the information carried in each.
4. **Plexuses:** cervical, brachial, lumbar, sacral.

Brain, meninges, and CSF

1. **Cerebrum:** gyri, sulci, longitudinal fissure, pre- and post-central gyri and sulci, central sulcus, lateral sulcus, lobes (frontal and occipital, unpaired; parietal and temporal, paired), insula, corpus callosum, cingulate gyrus and sulcus, septum pellucidum, fornix, uncus, parahippocampal gyrus, thalamus, massa intermedia (interthalamic adhesion), hypothalamus, infundibulum, pineal gland, optic chiasm, mammillary bodies, crus cerebri, transverse fissure (separates the cerebrum from the cerebellum).
2. **Brainstem and cerebellum:** pontomedullary junction, decussation of pyramids, medullary olive, midbrain, basilar pons, medulla oblongata, cerebellum.
3. **Functional areas:** Wernicke's area, Broca's area, primary sensory cortex, primary motor cortex, thalamus.
4. **Dural extensions:** falx cerebri, falx cerebelli, tentorium cerebelli (know what each separates).
5. **CSF and ventricles:** choroid plexus (with ependymal cells), lateral ventricles, interventricular foramen (of Monro), third ventricle, cerebral aqueduct, fourth ventricle; be able to trace CSF from the lateral ventricles to reabsorption (median and lateral apertures, arachnoid granulations).

Cranial nerves

1. Identify cranial nerves I-XII; classify each as sensory, motor, or both; know the function of each.
2. **Associated structures:** olfactory bulb and tract, olfactory foramina in the cribriform plate, optic nerve / tract / chiasm, superior orbital fissure, optic canal, internal acoustic meatus, pontomedullary junction.

Peripheral nerves by region

UPPER EXTREMITY

musculocutaneous nerve, median nerve, ulnar nerve, radial nerve (radial: triceps brachii heads, brachioradialis, the extensor compartment of the forearm, and skin of the dorsum of the hand).

LOWER EXTREMITY

femoral nerve (lumbar plexus); sciatic nerve (sacral plexus), branching into the tibial nerve and the common fibular (peroneal) nerve.

HEAD AND NECK

vagus nerve (CN X); facial nerve (CN VII) branches — temporal, zygomatic, buccal, marginal mandibular, cervical; trigeminal nerve (CN V) divisions — ophthalmic (V1), maxillary (V2), mandibular (V3).

THORAX / DIAPHRAGM

phrenic nerve (C3-C5), the innervation of the diaphragm.

7 · Endocrine System

1. Pituitary gland (in the hypophyseal fossa), pineal gland, thyroid gland, suprarenal (adrenal) glands, pancreas (endocrine islets). Identify each where it appears on the regional models.

8 · Cardiovascular System

Blood (formed elements)

1. **Peripheral blood smear:** leukocytes — neutrophils, basophils, eosinophils, monocytes, lymphocytes (and function); erythrocytes (and function); thrombocytes (and function).

Blood vessel histology

1. **Artery:** tunica intima (endothelium, internal elastic lamina), tunica media (smooth muscle, elastic CT), tunica externa, lumen, erythrocytes.
2. **Vein:** tunica intima (endothelium), tunica media (smooth muscle), tunica externa, lumen, erythrocytes.

The heart

1. **Coverings and wall:** fibrous pericardium, serous pericardium (parietal and visceral), epicardium, myocardium, endocardium.
2. **Atria:** right and left atria, right and left auricles, pectinate muscles, interatrial septum, fossa ovalis (remnant of the foramen ovale), SA node, AV node, coronary sinus, superior and inferior venae cavae.
3. **Ventricles:** right and left ventricles, interventricular septum, tricuspid (right AV) valve, pulmonary semilunar valve, mitral/bicuspid (left AV) valve, aortic semilunar valve, chordae tendineae, papillary muscles, trabeculae carneae.
4. **Coronary supply:** right coronary artery, left coronary artery, anterior interventricular artery (LAD), circumflex artery.
5. **Conduction system:** SA node, AV node, bundle of His, right and left bundle branches, Purkinje fibers (know intrinsic rates and trace the impulse).
6. **Great vessels:** ascending aorta, aortic arch and its three branches (brachiocephalic trunk, left common carotid, left subclavian), descending aorta, pulmonary trunk and right/left pulmonary arteries, four pulmonary veins.
7. **Fetal remnants:** fossa ovalis (foramen ovale), ligamentum arteriosum (ductus arteriosus). Trace a drop of blood through the pulmonary circuit.

Arteries by region

HEAD AND NECK

common carotid, internal carotid, external carotid, vertebral artery.

THORAX

ascending aorta, aortic arch, descending (thoracic) aorta, brachiocephalic trunk, left common carotid, left subclavian, pulmonary trunk and pulmonary arteries.

UPPER EXTREMITY

subclavian, axillary, brachial (and the radial and ulnar arteries of the forearm).

ABDOMEN

abdominal aorta; celiac trunk and branches (left gastric, splenic, common hepatic); superior mesenteric; inferior mesenteric; renal; gonadal (testicular / ovarian); right and left common iliac.

PELVIS

internal iliac, external iliac; medial umbilical ligament (remnant of the umbilical arteries).

LOWER EXTREMITY

femoral artery, posterior tibial artery.

BRAIN (CIRCLE OF WILLIS)

internal carotid, anterior cerebral, middle cerebral, posterior cerebral, anterior and posterior communicating, basilar, vertebral, pontine, superior cerebellar, anterior and posterior inferior cerebellar, anterior spinal.

KIDNEY (INTRINSIC)

renal, arcuate, cortical radiate, afferent and efferent arterioles, peritubular capillaries, vasa recta.

Veins by region**HEAD AND NECK**

internal jugular vein.

THORAX / GREAT VEINS

superior vena cava, inferior vena cava, right/left brachiocephalic, right/left subclavian, right/left internal jugular, coronary sinus, pulmonary veins (all systemic veins listed empty into the right atrium).

UPPER EXTREMITY

deep — subclavian, axillary, brachial; superficial — basilic, cephalic, median cubital.

ABDOMEN

inferior vena cava and right/left common iliac veins; hepatic portal system — hepatic portal vein, splenic vein, superior and inferior mesenteric veins.

PELVIS

internal iliac, external iliac.

LOWER EXTREMITY

femoral vein (deep), great saphenous vein (superficial).

KIDNEY (INTRINSIC)

renal, arcuate, cortical radiate veins.

BRAIN

dural venous sinuses — superior and inferior sagittal, straight, sigmoid, confluence of sinuses.

9 · Lymphatic System

1. Spleen; tonsils (pharyngeal, palatine, lingual).

10 · Respiratory System

1. **Larynx:** thyroid cartilage, laryngeal prominence, cricoid cartilage, cricothyroid ligament, arytenoid cartilages, vocal cords (folds), glottis, epiglottis.
2. **Trachea and bronchi:** tracheal cartilages (C-rings), trachealis muscle, carina, right/left primary (main) bronchi, secondary (lobar) bronchi, tertiary (segmental) bronchi, alveolar duct, alveoli.
3. **Lungs and pleura:** right lung lobes (superior, middle, inferior), left lung lobes (superior, inferior), hilum, parietal and visceral pleura, pleural cavities.
4. **Thoracic wall and diaphragm:** diaphragm, external and internal intercostals; the three structures passing through the diaphragm and their levels (T8, T10, T12).

11 · Digestive System

Abdominal blood vessels are listed once, under Cardiovascular → Arteries/Veins by region (Abdomen). They are not repeated here.

1. **Peritoneum and omenta:** abdominal cavity, pelvic cavity, peritoneal cavity, parietal and visceral peritoneum, retroperitoneum, greater omentum, lesser omentum.
2. **GI tract — stomach:** greater curvature, lesser curvature, gastric rugae, cardiac region, fundus, body, pyloric region, pyloric sphincter, cardiac sphincter. Esophagus; submandibular (salivary) glands.
3. **Small intestine (small bowel):** mesentery, duodenum, major duodenal papilla, jejunum, ileum, ileocecal valve.
4. **Large intestine (large bowel):** cecum, vermiform appendix, ascending colon, hepatic (right colic) flexure, transverse colon, splenic (left colic) flexure, descending colon, sigmoid colon, rectum, teniae coli, haustra, anal canal.
5. **Accessory organs — pancreas:** main and accessory pancreatic ducts, hepatopancreatic ampulla, major duodenal papilla. **Gallbladder. Biliary tree:** right and left hepatic ducts, common hepatic duct, cystic duct, common bile duct. **Liver:** lobes (right, left, quadrate, caudate), falciform ligament, ligamentum teres (remnant of the umbilical vein).

12 · Urinary System

1. **Gross:** right and left kidney, suprarenal (adrenal) gland, renal capsule, renal cortex, renal medulla, renal column, renal pyramid, minor calyx, major calyx, renal pelvis, renal sinus, ureter, urinary bladder, urethra.
2. **Renal corpuscle:** glomerulus, glomerular (Bowman's) capsule with parietal and visceral layers.
3. **Renal tubule:** proximal convoluted tubule, loop of Henle, distal convoluted tubule, collecting duct, papillary duct, juxtaglomerular apparatus (macula densa, granular cells). Trace filtrate from the PCT to outside the body, and blood through the kidney. (Renal vasculature is under Cardiovascular → Arteries/Veins by region, Kidney.)

13 · Reproductive System

1. **Male:** scrotum, testes, spermatic cord, ductus (vas) deferens, testicular artery and vein, cremaster muscle, pampiniform plexus, inguinal canal, ejaculatory duct, seminal vesicles, prostate gland, bulbourethral glands, penis, corpus spongiosum, corpora cavernosa, glans penis, prepuce, urethra (three regions: prostatic urethra, membranous urethra, spongy or penile urethra). Trace sperm from the seminiferous tubules to the outside.
2. **Female:** ovary, ovarian ligament, ovarian artery and vein, broad ligament, suspensory ligament, uterine (fallopian) tubes / oviduct (fimbriae, infundibulum, ampulla, isthmus), round ligament of the uterus, uterus (fundus, body,

cervix, endometrium, myometrium, perimetrium), vagina (orifice, fornix), vulva (mons pubis, labia majora, labia minora, clitoris, urethral orifice).

3. **Pelvic vessels:** see Cardiovascular → Arteries/Veins by region (Pelvis): internal and external iliac arteries and veins; medial umbilical ligament (remnant of the umbilical artery).